



Habib University
shaping futures

Instructions

- The exam will begin at **5:40 PM** and conclude at **7:10 PM**, with a total duration of **1 hour and 30 minutes**. The exam is worth **30 points** in total.
- This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
- Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the class room before the start of the test.
- You are to attempt all question on the answer sheet provided. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
- Use of calculators is allowed for this exam. Any form of communication/sharing among your peers is not allowed during the exam.
- If you want to communicate with the invigilator, raise your hand.
- You are not allowed to leave your desk without permission of the invigilator during the exam.
- Failure to abide by the instructions above would lead to immediate cancellation of the exam.

Student Name: _____

Instructor's Name: _____

Section: _____

1. (4 points) Find the mass of a wire in the shape of a helix

$$x = t, \quad y = \cos t, \quad z = \sin t, \quad 0 \leq t \leq 2\pi,$$

if the density at any point is equal to the square of the distance from the origin.

Solution:

The helix is parametrized by

$$\mathbf{r}(t) = (x(t), y(t), z(t)) = (t, \cos t, \sin t), \quad 0 \leq t \leq 2\pi.$$

The density at a point equals the square of the distance from the origin:

$$\rho(t) = x^2 + y^2 + z^2 = t^2 + \cos^2 t + \sin^2 t = t^2 + 1.$$

The speed is

$$\mathbf{r}'(t) = (1, -\sin t, \cos t), \quad \|\mathbf{r}'(t)\| = \sqrt{1 + \sin^2 t + \cos^2 t} = \sqrt{2}.$$

Mass

The mass is

$$m = \int_0^{2\pi} \rho(t) \|\mathbf{r}'(t)\| dt = \sqrt{2} \int_0^{2\pi} (t^2 + 1) dt = \sqrt{2} \left[\frac{t^3}{3} + t \right]_0^{2\pi}.$$

Hence

$$m = \sqrt{2} \left(\frac{(2\pi)^3}{3} + 2\pi \right) = \frac{2\sqrt{2}\pi}{3} (3 + 4\pi^2).$$

2. (5 points) Suppose that $\vec{F} = \vec{\nabla} f$ is a conservative vector field and

$$g(x, y, z) = \int_{(0,0,0)}^{(x,y,z)} \vec{F} \cdot d\vec{r}.$$

Show that $\vec{\nabla} g = \vec{F}$.

Solution:

Let $\vec{\mathbf{F}} = \vec{\nabla} f$ be a conservative vector field on a region containing the origin, and define

$$g(x, y, z) = \int_{(0,0,0)}^{(x,y,z)} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}},$$

where $C_0^{(x,y,z)}$ is any smooth curve from $(0,0,0)$ to (x,y,z) . Because $\vec{\mathbf{F}}$ is conservative the integral is path independent, so g is well defined.

Since $\vec{\mathbf{F}} = \vec{\nabla} f$, the fundamental theorem for line integrals gives, for every (x, y, z) ,

$$g(x, y, z) = \int_{(0,0,0)}^{(x,y,z)} \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(x, y, z) - f(0, 0, 0).$$

Hence $g = f - f(0, 0, 0)$. Differentiating both sides yields

$$\vec{\nabla} g = \vec{\nabla} f = \vec{\mathbf{F}}.$$

Therefore $\vec{\nabla} g = \vec{\mathbf{F}}$, as required.

3. (6 points) Consider the vector field

$$\vec{\mathbf{F}}(x, y) = \langle -y, x \rangle.$$

(a) Compute the line integral

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}},$$

where C is the quarter-circle of radius 2 in the first quadrant, traversed counterclockwise from $(2, 0)$ to $(0, 2)$.

(b) Compute the same integral, but now let C consist of two straight line segments: from $(2, 0)$ to $(0, 0)$, and then from $(0, 0)$ to $(0, 2)$.

(c) Compare your answers from (a) and (b). Why are they different? What does this tell you about the vector field $\vec{\mathbf{F}}$?

Solution:

Solution

Let $\vec{\mathbf{F}}(x, y) = \langle -y, x \rangle$. Write $\vec{\mathbf{F}} = (P, Q)$ with $P(x, y) = -y$ and $Q(x, y) = x$.

(a) Quarter-circle arc.

Parametrize the quarter-circle of radius 2 in the first quadrant, counterclockwise from $(2, 0)$ to $(0, 2)$, by

$$\vec{r}(t) = \langle 2 \cos t, 2 \sin t \rangle, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Then

$$\vec{r}'(t) = \langle -2 \sin t, 2 \cos t \rangle, \quad \vec{F}(\vec{r}(t)) = \langle -2 \sin t, 2 \cos t \rangle.$$

Hence the integrand is

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = (-2 \sin t)(-2 \sin t) + (2 \cos t)(2 \cos t) = 4(\sin^2 t + \cos^2 t) = 4.$$

Therefore

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} 4 \, dt = 4 \cdot \frac{\pi}{2} = 2\pi.$$

(b) Two straight segments via the origin.

We take $C = C_1 \cup C_2$ where C_1 is the segment from $(2, 0)$ to $(0, 0)$ and C_2 is the segment from $(0, 0)$ to $(0, 2)$.

Use the parametrizations (with $t \in [0, 1]$):

$$\vec{r}_1(t) = \langle 2 - 2t, 0 \rangle, \quad 0 \leq t \leq 1,$$

so $\vec{r}_1'(t) = \langle -2, 0 \rangle$. For this curve

$$\vec{F}(\vec{r}_1(t)) = \langle -0, 2 - 2t \rangle = \langle 0, 2 - 2t \rangle.$$

Thus

$$\vec{F}(\vec{r}_1(t)) \cdot \vec{r}_1'(t) = 0 \cdot (-2) + (2 - 2t) \cdot 0 = 0,$$

and $\int_{C_1} \vec{F} \cdot d\vec{r} = 0$.

For C_2 take

$$\vec{r}_2(t) = \langle 0, 2t \rangle, \quad 0 \leq t \leq 1,$$

so $\vec{r}_2'(t) = \langle 0, 2 \rangle$. For this curve

$$\vec{F}(\vec{r}_2(t)) = \langle -2t, 0 \rangle,$$

and

$$\vec{F}(\vec{r}_2(t)) \cdot \vec{r}_2'(t) = (-2t) \cdot 0 + 0 \cdot 2 = 0,$$

so $\int_{C_2} \vec{F} \cdot d\vec{r} = 0$.

Therefore for the two-segment path

$$\int_C \vec{\mathbf{F}} \cdot d\vec{r} = \int_{C_1} + \int_{C_2} = 0 + 0 = 0.$$

(c) Comparison and interpretation.

From parts (a) and (b) we have

$$\int_{\text{arc}} \vec{\mathbf{F}} \cdot d\vec{r} = 2\pi, \quad \int_{\text{segments}} \vec{\mathbf{F}} \cdot d\vec{r} = 0.$$

They are different, so the line integral depends on the path between the same endpoints.

A direct test shows the field is not conservative: $P_y = \partial(-y)/\partial y = -1$ while $Q_x = \partial x/\partial x = 1$, so $P_y \neq Q_x$. Thus $\vec{\mathbf{F}}$ is not conservative (equivalently, the planar curl is $2 \neq 0$). The inequality of mixed partials alone is sufficient to conclude non-conservativeness here; one does not need to appeal to domain topology for this example.

Answers: (a) 2π , (b) 0 , (c) $\vec{\mathbf{F}}$ is not conservative ($P_y \neq Q_x$).

4. (7 points) The force exerted by an electric charge at the origin on a charged particle at a point (x, y, z) with position vector $\vec{r} = \langle x, y, z \rangle$ is

$$\vec{F}(\vec{r}) = \frac{K\vec{r}}{|\vec{r}|^3},$$

where K is a constant and $\{(x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}\}$. Find the work done as the particle moves along a straight line from $(2, 0, 0)$ to $(2, 1, 5)$.

Solution: Let

$$\mathbf{F}(x, y, z) = \frac{K\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}} = \langle F_1, F_2, F_3 \rangle,$$

where

$$F_1(x, y, z) = \frac{Kx}{(x^2 + y^2 + z^2)^{3/2}}, \quad F_2(x, y, z) = \frac{Ky}{(x^2 + y^2 + z^2)^{3/2}}$$

, and

$$F_3(x, y, z) = \frac{Kz}{(x^2 + y^2 + z^2)^{3/2}}.$$

1. Check cross partials).

Set $s = x^2 + y^2 + z^2$. Compute partials:

$$\frac{\partial F_1}{\partial y} = K \frac{\partial}{\partial y}(xs^{-3/2}) = Kx \cdot \left(-\frac{3}{2}\right) s^{-5/2} \cdot 2y = -3Kxy s^{-5/2},$$

and

$$\frac{\partial F_2}{\partial x} = Ky \cdot \left(-\frac{3}{2}\right) s^{-5/2} \cdot 2x = -3Kxy s^{-5/2}.$$

Hence $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$.

Similarly,

$$\frac{\partial F_1}{\partial z} = -3Kxz s^{-5/2}, \quad \frac{\partial F_3}{\partial x} = -3Kxz s^{-5/2},$$

and

$$\frac{\partial F_2}{\partial z} = -3Kyz s^{-5/2}, \quad \frac{\partial F_3}{\partial y} = -3Kyz s^{-5/2}.$$

Thus all cross partial equalities hold and $\vec{\nabla} \times \vec{\mathbf{F}} = \mathbf{0}$ on $\mathbb{R}^3 \setminus \{0\}$.

Because the domain $\mathbb{R}^3 \setminus \{0\}$ is simply connected, $\mathbf{F}$ is conservative there and admits a scalar potential ϕ with $\mathbf{F} = \nabla \phi$.

2. Find the potential function.

We seek ϕ satisfying

$$\phi_x = \frac{Kx}{(x^2 + y^2 + z^2)^{3/2}}.$$

Integrate with respect to x (treat y, z as constants). Let $u = x^2 + y^2 + z^2$, so $du = 2x dx$. Then

$$\int \frac{Kx}{u^{3/2}} dx = \frac{K}{2} \int u^{-3/2} du = \frac{K}{2} \cdot (-2u^{-1/2}) + g(y, z) = -\frac{K}{\sqrt{u}} + g(y, z).$$

Thus

$$\phi(x, y, z) = -\frac{K}{\sqrt{x^2 + y^2 + z^2}} + g(y, z).$$

Differentiate this with respect to y :

$$\phi_y = Ky(x^2 + y^2 + z^2)^{-3/2} + g_y(y, z).$$

Equating to $F_2 = Ky(x^2 + y^2 + z^2)^{-3/2}$ gives $g_y = 0$. Similarly $g_z = 0$. Therefore g is constant and we may take

$$\boxed{\phi(x, y, z) = -\frac{K}{\sqrt{x^2 + y^2 + z^2}} + C}.$$

3. Work from $A = (2, 0, 0)$ to $B = (2, 1, 5)$ by the Fundamental Theorem of Line Integrals.

Since $\mathbf{F} = \nabla\phi$ on any path avoiding the origin, for any smooth curve C from A to B ,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A).$$

Compute the potentials at the endpoints (take $C = 0$ for simplicity):

$$|A| = \sqrt{2^2 + 0^2 + 0^2} = 2, \quad \phi(A) = -\frac{K}{2} + C,$$

$$|B| = \sqrt{2^2 + 1^2 + 5^2} = \sqrt{30}, \quad \phi(B) = -\frac{K}{\sqrt{30}} + C.$$

Therefore the work is

$$W = \phi(B) - \phi(A) = \left(-\frac{K}{\sqrt{30}} + C\right) - \left(-\frac{K}{2} + C\right) = K\left(\frac{1}{2} - \frac{1}{\sqrt{30}}\right).$$

Hence

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = K\left(\frac{1}{2} - \frac{1}{\sqrt{30}}\right).}$$

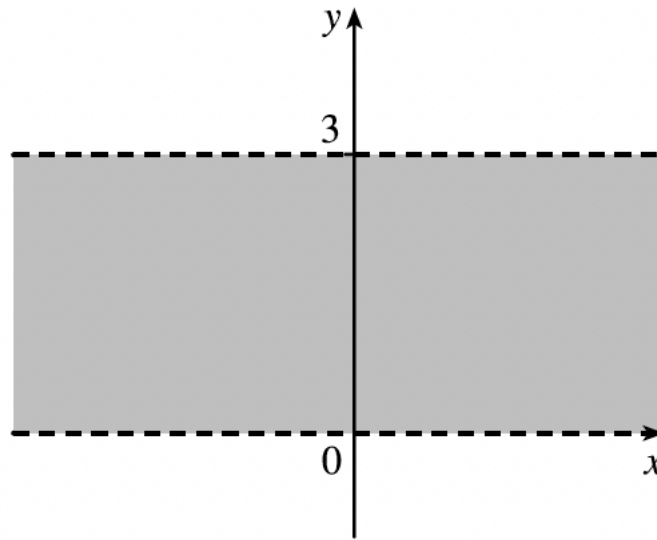
5. (8 points) Determine whether or not the given set is (a) open, (b) connected, and (c) simply connected.

- $\{(x, y) \mid 0 < y < 3\}$
- $\{(x, y) \mid 1 < |x| < 2\}$
- $\{(x, y) \mid 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$
- $\{(x, y) \mid (x, y) \neq (2, 3)\}$

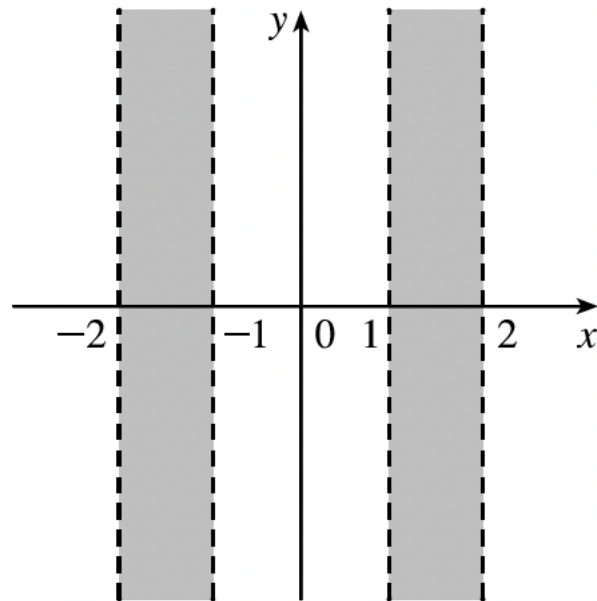
Solution:

1. Let $S_1 = \{(x, y) \mid 0 < y < 3\}$ consists of those points between, but not on, the horizontal lines $y = 0$ and $y = 3$.
 - (a) Since S_1 does not include any of its boundary points, it is open. More formally, at any point in S_1 there exists a disk centered at that point that lies entirely within S_1 .

- (b) Any two points in S_1 can be joined by a path that lies entirely within S_1 . Therefore, S_1 is connected. (In fact, S_1 consists of just one “piece.”)
- (c) S_1 is connected and it has no holes, so it’s simply-connected. (Every simple closed curve in S_1 encloses only points that are in S_1 .)

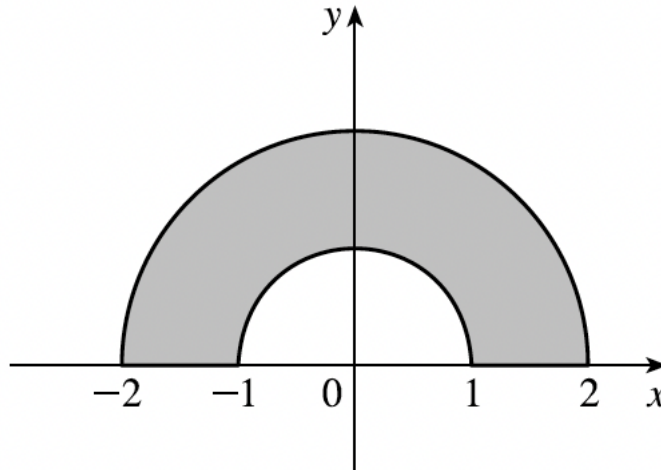


2. Let $S_2 = \{(x, y) \mid 1 < |x| < 2\}$ consists of those points between, but not on, the vertical lines $x = 1$ and $x = 2$, together with the points between the vertical lines $x = -1$ and $x = -2$.
- (a) The region does not include any of its boundary points, so it is open.
 - (b) S_2 consists of two separate pieces, so it is not connected. For instance, the points $(-1.5, 0)$ and $(1.5, 0)$ both lie in S_2 , but they cannot be joined by a path that stays entirely within S_2 .
 - (c) Because S_2 is not connected, it is not simply connected.



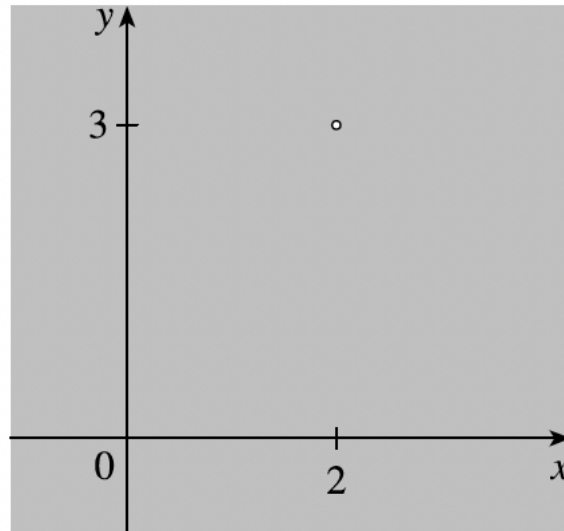
3. Let $S_3 = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$ is the semi-annular region in the upper half-plane between circles centered at the origin of radii 1 and 2 (including all boundary points).

- (a) S_3 includes boundary points, so it is not open. For instance, at a boundary point such as $(1, 0)$, any disk centered there cannot lie entirely in S_3 .
- (b) The region consists of one piece, so it is connected.
- (c) S_3 is connected and has no holes, so it is simply connected.



4. Let $S_4 = \{(x, y) \mid (x, y) \neq (2, 3)\}$ consists of all points in the xy -plane except for $(2, 3)$.

- (a) S_4 has only one boundary point, namely $(2, 3)$, which is not included, so the region is open.
- (b) S_4 is connected, as it consists of only one piece.
- (c) S_4 is not simply connected, as it has a hole at $(2, 3)$. Thus, any simple closed curve that encloses $(2, 3)$ lies in S_4 but contains a point that is not in S_4 .



Important Formulae

Concept	Formula
Del Operator	$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$
Gradient of a scalar function $\phi(x, y, z)$	$\vec{\nabla} \phi$
Divergence of a vector field $\vec{F}$	$\vec{\nabla} \cdot \vec{F}$
Curl of a vector field $\vec{F}$	$\vec{\nabla} \times \vec{F}$
Scalar line integral	$\int_C f(x, y, z) ds = \int_C f(\vec{r}(t)) \vec{r}'(t) dt$
Mass of a wire (1D curve C)	$M = \int_C \rho(x, y, z) ds$
Center of Mass (x -coordinate)	$\bar{x} = \frac{1}{M} \int_C x \rho(x, y, z) ds$
Center of Mass (y -coordinate)	$\bar{y} = \frac{1}{M} \int_C y \rho(x, y, z) ds$
Center of Mass (z -coordinate)	$\bar{z} = \frac{1}{M} \int_C z \rho(x, y, z) ds$
Line integral of a vector field	$\int_C (\vec{F} \cdot \vec{T}) ds = \int_C \vec{F} \cdot d\vec{r}$
Closed line integral	$\oint_C \vec{F} \cdot d\vec{r}$
Flux of a vector field in 2D	$\int_C (\vec{F} \cdot \vec{N}) ds = \oint M dy - N dx$
For conservative field $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$	$\vec{F} = \vec{\nabla} f$ for some scalar function f
Path-independence (between (x_0, y_0, z_0) and (x_1, y_1, z_1))	$\int_C \vec{F} \cdot d\vec{r} = \int_{C'} \vec{F} \cdot d\vec{r}$
Closed curve integral (conservative field)	$\oint_C \vec{F} \cdot d\vec{r} = 0$
Conservative field test for $\vec{F} = \langle F_x, F_y, F_z \rangle$	$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}, \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}$
Fundamental theorem of line integral	$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{\nabla} f \cdot d\vec{r} = f(x_1, y_1, z_1) - f(x_0, y_0, z_0)$